

$$q_{1,1,2,3,14} \rightarrow$$

$$\begin{aligned} & \left(\frac{1}{16\pi^2} \left(\frac{1}{2592} \frac{1}{m_s^2} \left(s_Y^2 \overline{y}_d^{i4p} (y_d^{i3p} (162 c_Y^2 \overline{y}_u^{i2r} y_u^{i1r} + (5 g_1^2 + 24 g_3^2) \delta_{i1i2}) - 36 g_3^2 y_d^{i1p} \delta_{i2i3} \right) + \right. \right. \\ & s_Y^2 \overline{y}_d^{i2p} (-36 g_3^2 y_d^{i3p} \delta_{i1i4} + \\ & y_d^{i1p} (162 c_Y^2 (-\overline{y}_d^{i4r} y_d^{i3r} + \overline{y}_u^{i4r} y_u^{i3r}) + (5 g_1^2 + 24 g_3^2) \delta_{i3i4})) + \\ & c_Y^2 (\overline{y}_u^{i4p} (y_u^{i3p} (-13 g_1^2 + 24 g_3^2) \delta_{i1i2} - 36 g_3^2 y_u^{i1p} \delta_{i2i3}) + \\ & \overline{y}_u^{i2p} (-36 g_3^2 y_u^{i3p} \delta_{i1i4} + y_u^{i1p} (-162 s_Y^2 \overline{y}_u^{i4r} y_u^{i3r} + (-13 g_1^2 + 24 g_3^2) \delta_{i3i4}))) \Big) \Big) + \\ & \frac{1}{36} g_3^4 (3 \delta_{i1i4} \delta_{i2i3} - 2 \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[\mathbf{m}_3] + \frac{1}{24} g_3^4 (3 \delta_{i1i4} \delta_{i2i3} - 2 \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[\mathbf{m}_3] + \\ & \frac{2}{45} g_3^4 (-3 \delta_{i1i4} \delta_{i2i3} + 2 \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[\mathbf{m}_3] + \\ & \frac{1}{972} \sum_p (27 g_3^4 \delta_{i1i4} \delta_{i2i3} + 2 (g_1^4 - 9 g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[\mathbf{m}_d^p] - \\ & \frac{5}{2592} \sum_p (27 g_3^4 \delta_{i1i4} \delta_{i2i3} + 2 (g_1^4 - 9 g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[\mathbf{m}_d^p] + \\ & \frac{1}{1215} \sum_p (27 g_3^4 \delta_{i1i4} \delta_{i2i3} + 2 (g_1^4 - 9 g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[\mathbf{m}_d^p] + \\ & \frac{1}{162} \sum_p g_1^4 \text{LF}_{3,0}[\mathbf{m}_e^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{432} \sum_p g_1^4 \text{LF}_{4,-1}[\mathbf{m}_e^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{405} \sum_p g_1^4 \text{LF}_{5,-2}[\mathbf{m}_e^p] \delta_{i1i2} \delta_{i3i4} + \frac{1}{324} \sum_p g_1^4 \text{LF}_{3,0}[\mathbf{m}_l^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{864} \sum_p g_1^4 \text{LF}_{4,-1}[\mathbf{m}_l^p] \delta_{i1i2} \delta_{i3i4} + \frac{1}{405} \sum_p g_1^4 \text{LF}_{5,-2}[\mathbf{m}_l^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{972} \sum_p (54 g_3^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - 36 g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[\mathbf{m}_q^p] - \\ & \frac{5}{2592} \sum_p (54 g_3^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - 36 g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[\mathbf{m}_q^p] + \\ & \frac{1}{1215} \sum_p (54 g_3^4 \delta_{i1i4} \delta_{i2i3} + (g_1^4 - 36 g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[\mathbf{m}_q^p] + \\ & \frac{1}{972} \sum_p (27 g_3^4 \delta_{i1i4} \delta_{i2i3} + 2 (4 g_1^4 - 9 g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,0}[\mathbf{m}_u^p] - \\ & \frac{5}{2592} \sum_p (27 g_3^4 \delta_{i1i4} \delta_{i2i3} + 2 (4 g_1^4 - 9 g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,-1}[\mathbf{m}_u^p] + \\ & \frac{1}{1215} \sum_p (27 g_3^4 \delta_{i1i4} \delta_{i2i3} + 2 (4 g_1^4 - 9 g_3^4) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{5,-2}[\mathbf{m}_u^p] + \\ & \frac{1}{216} (s_Y^2 \overline{y}_d^{i4p} (y_d^{i3p} (27 c_Y^2 \overline{y}_u^{i2r} y_u^{i1r} + 2 (g_1^2 + 3 g_3^2) \delta_{i1i2}) - 9 g_3^2 y_d^{i1p} \delta_{i2i3}) + s_Y^2 \overline{y}_d^{i2p} \\ & (-9 g_3^2 y_d^{i3p} \delta_{i1i4} + y_d^{i1p} (27 c_Y^2 (\overline{y}_d^{i4r} y_d^{i3r} + \overline{y}_u^{i4r} y_u^{i3r}) + 2 (g_1^2 + 3 g_3^2) \delta_{i3i4})) + \\ & c_Y^2 (\overline{y}_u^{i4p} (2 y_u^{i3p} (-2 g_1^2 + 3 g_3^2) \delta_{i1i2} - 9 g_3^2 y_u^{i1p} \delta_{i2i3}) + \\ & \overline{y}_u^{i2p} (-9 g_3^2 y_u^{i3p} \delta_{i1i4} + y_u^{i1p} (27 s_Y^2 \overline{y}_u^{i4r} y_u^{i3r} + 2 (-2 g_1^2 + 3 g_3^2) \delta_{i3i4}))) \Big) \Big) \\ & \text{LF}_{1,2}[\mathbf{m}_\Phi] + \frac{1}{144} (-s_Y^2 \overline{y}_d^{i4p} y_d^{i3p} (9 c_Y^2 \overline{y}_u^{i2r} y_u^{i1r} + g_1^2 \delta_{i1i2}) + \\ & s_Y^2 \overline{y}_d^{i2p} y_d^{i1p} (9 s_Y^2 \overline{y}_d^{i4r} y_d^{i3r} - 9 c_Y^2 \overline{y}_u^{i4r} y_u^{i3r} - g_1^2 \delta_{i3i4}) + \\ & c_Y^2 (g_1^2 \overline{y}_u^{i4p} y_u^{i3p} \delta_{i1i2} + \overline{y}_u^{i2p} y_u^{i1p} (9 c_Y^2 \overline{y}_u^{i4r} y_u^{i3r} + g_1^2 \delta_{i3i4}))) \text{LF}_{2,1}[\mathbf{m}_\Phi] + \\ & \frac{1}{1296} g_1^2 (9 (s_Y^2 \overline{y}_d^{i4p} y_d^{i3p} - c_Y^2 \overline{y}_u^{i4p} y_u^{i3p}) \delta_{i1i2} + \\ & (9 s_Y^2 \overline{y}_d^{i2p} y_d^{i1p} - 9 c_Y^2 \overline{y}_u^{i2p} y_u^{i1p} + 4 g_1^2 \delta_{i1i2}) \delta_{i3i4}) \text{LF}_{3,0}[\mathbf{m}_\Phi] - \\ & \frac{5}{864} g_1^4 \text{LF}_{4,-1}[\mathbf{m}_\Phi] \delta_{i1i2} \delta_{i3i4} + \frac{1}{405} g_1^4 \text{LF}_{5,-2}[\mathbf{m}_\Phi] \delta_{i1i2} \delta_{i3i4} + \frac{1}{324} g_1^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{216} g_1^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{405} g_1^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{15552} g_1^2 (9 g_3^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - 6 g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_q^{i1}] + \\ & \frac{1}{15552} g_1^2 (9 g_3^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - 6 g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}] + \\ & \frac{1}{7776} g_1^2 (-9 g_3^2 \delta_{i1i4} \delta_{i2i3} - (g_1^2 - 6 g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}] + \\ & \frac{1}{15552} g_1^2 (9 g_3^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - 6 g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,1,-2}[\mathbf{m}_1, \mathbf{m}_q^{i1}] + \\ & \frac{1}{15552} g_1^2 (9 g_3^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - 6 g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_q^{i2}] + \\ & \frac{1}{15552} g_1^2 (9 g_3^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - 6 g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[\mathbf{m}_1, \mathbf{m}_q^{i2}] + \\ & \frac{1}{7776} g_1^2 (-9 g_3^2 \delta_{i1i4} \delta_{i2i3} - (g_1^2 - 6 g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{3,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i2}] + \\ & \frac{1}{15552} g_1^2 (9 g_3^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - 6 g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{4,1,-2}[\mathbf{m}_1, \mathbf{m}_q^{i2}] + \\ & \frac{1}{15552} g_1^2 (9 g_3^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - 6 g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_q^{i3}] + \\ & \frac{1}{15552} g_1^2 (9 g_3^2 \delta_{i1i4} \delta_{i2i3} + (g_1^2 - 6 g_3^2) \delta_{i1i2} \delta_{i3i4}) \text{LF}_{2,2,-1}[\mathbf{m}_1$$